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Evaluating the relationship between magnetic resonance image quality metrics and deep learning-based segmentation accuracy of brain tumors

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Abstract

Background: Magnetic resonance imaging (MRI) scans are known to suffer from a variety of acquisition artifacts as well as equipment-based variations that impact image appearance and segmentation performance. It is still unclear whether a direct relationship exists between MR image quality metrics (IQMs) (e.g., signal-to-noise, contrast-to-noise) and segmentation accuracy.

Purpose: Deep learning approaches have shown significant promise for automated segmentation of brain tumors on MRI but depend on the quality of input training images. We sought to evaluate the relationship between IQMs of input training images and deep learning-based brain tumor segmentation accuracy toward developing more generalizable models for multi-institutional data.

Methods: We trained a 3D DenseNet model on the BraTS 2020 cohorts for segmentation of tumor subregions enhancing tumor, peritumoral edematous, and necrotic and non-enhancing tumor on MRI; with performance quantified via a 5-fold cross-validated Dice coefficient. MRI scans were evaluated through the open-source quality control tool MRQy, to yield 13 image quality measures (IQMs) per scan. The Pearson correlation coefficient was computed between whole tumor (WT) dice values and IQM measures in the training cohorts to identify quality measures

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CONFLICT OF INTEREST STATEMENT

The authors have no relevant conflicts of interest to disclose.

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most correlated with segmentation performance. Each selected IQM was used to group MRI scans as “better” quality (BQ) or “worse” quality (WQ), via relative thresholding. Segmentation performance was re-evaluated for the DenseNet model when (i) training on BQ MRI images with validation on WQ images, as well as (ii) training on WQ images, and validation on BQ images. Trends were further validated on independent test sets derived from the BraTS 2021 training cohorts.

Results: For this study, multimodal MRI scans from the BraTS 2020 training cohorts were used to train the segmentation model and validated on independent test sets derived from the BraTS 2021 cohort. Among the selected IQMs, models trained on BQ images based on inhomogeneity measurements (coefficient of variance (CV), coefficient of joint variation (CJV), coefficient of variation of the foreground patch (CVP)) and the models trained on WQ images based on noise measurement peak SNR (PSNR)) yielded significantly improved tumor segmentation accuracy compared to their inverse models.

Conclusions: Our results suggest that a significant correlation may exist between specific MR image quality measures and DenseNet-based brain tumor segmentation performance. The selection of MRI scans for model training based on IQMs may yield more accurate and generalizable models in unseen validation.

1 INTRODUCTION

The success of deep learning (DL) based image segmentation tools for tumors and normal tissue structures could have significant potential to improve workflow efficiencies in clinical trials as well as clinical practice[1]. In 3D medical image segmentation, the U-net architecture[2] with fully connected convolutional neural networks (CNNs) has shown significant potential in capturing both local and global features on radiographic images for multi-class segmentation[3–6]. More recently, dual pathway architectures with lateral connections (such as DenseNets) have shown improved performance in this regard, as the encoder extracts the features separately and then concatenates the features to the decoder[7,8]. For brain tumor segmentation, modality-specific information from multimodal brain magnetic resonance imaging (MRI) scans has been extracted through the multi-encoder architectures[9]. Fusing this complementary information of MR images from each modality to the higher-level layer has been shown to improve model performance[10–12].

Though various strategies have been implemented to improve the model performance, evaluation of the trained deep learning segmentation models onto new, unseen cohorts (especially in multi-center settings) has been highly challenging. A robust multi-center model created by combining structural brain MRI data reduced the risk of misclassification by finding effects common to all centers[13]. Very high classification accuracy has been reported using multi-center data for accurate detection of Alzheimer’s disease[14]. However, model performance has also been shown to vary significantly across variations in MR acquisition protocols and resolutions[15], with performance degradation due to inter-site heterogeneity when mixing multi-site data[16]. The presence of image quality distortions can pose a major limitation on applying the trained algorithms on a different dataset with varying image quality,[17,18] which may not be correctable via image normalization techniques alone[19].

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These reports suggest that the generalizability of deep learning models may be highly dependent on MR image quality and require more detailed knowledge of inherent variability within the training cohort. Recently several automated tools have been developed for evaluating variations and relative image quality measurements (IQMs)[20,21] in multi-center MRI cohorts. These have shown promise for automated quality control of MRIs on large multi-center datasets where visual inspection is unfeasible. This helps to control the overall quality of the images taken for model training which may regulate the neural network learning efficiency. However, it is still unclear whether a direct relationship exists between MR image quality measures (e.g., signal-to-noise, contrast-to-noise) and segmentation accuracy.

To develop a robust automated segmentation model, independent of dataset heterogeneity, the dataset can be curated before deployment either by imposing required correction to the dataset in terms of IQM measure or by selecting the data with respect to IQM measure for the specific model pipeline. We hypothesized that assessing the relative quality differences within the dataset could be utilized to reduce the impact of data variability in neural network training. The goal of the current study was to evaluate the relationship between deep learning-based brain tumor segmentation accuracy and MR IQMs in a multi-institutional setting.

2 METHODS

2.1 Data set

This study used publicly available data provided by Medical Image Computing and Computer Assisted Interventions Society's Multimodal Brain Tumor Image Segmentation Challenge: BraTS 2020[22] and BraTS 2021[23]. The database includes pre-operative multi-parametric MRI scans acquired from multiple institutions with different clinical protocols. The training cohort from BraTS 2020 comprised 369 patient datasets including T1-weighted (T1), post-contrast T1-weighted (T1Gd), T2-weighted (T2), and T2 Fluid Attenuated Inversion Recovery (T2-FLAIR) volumes. For each patient, annotations available include the necrotic and non-enhancing tumor core (NCR/NET-label 1), peritumoral edematous (ED-label 2), and enhancing tumor (ET-label 4). As released, all volumes have been standardized to a resolution of 240 × 240 × 155 after co-registration to the same anatomical template, interpolated to the same resolution (1 mm^3), and skull stripped.

From BraTS 2020 training dataset, 306 MRI scans were utilized for training and the remaining 63 scans were utilized as test set for 5-fold cross-validation. Independent validation cohorts were curated from the BraTS 2021 training dataset for IQM-based training scenarios (as explained in Section 2.4). In the BraTS 2021 training dataset (1666), we removed 369 cohorts from BraTS 2020 training dataset with mapping details available from BraTS website (to ensure no overlap between cohorts). From the remaining dataset, 70 cases were utilized to validate each training scenario. For pre-processing, we applied z-score intensity normalization for each modality independently for both the BraTS 2020 and BraTS 2021 dataset.

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2.2 Network Architecture and Training

A 3D DenseNet[24] was implemented for multiclass segmentation of ET, ED, and NCR/NET regions via brain MRIs. Four resolution levels were used. At a given resolution, the feature maps of preceding layers are used as inputs and produce an output feature map that is used as input by the subsequent layer which makes the network densely connected. The network was constructed as a combination of convolution and down sampling operations that extracted features along a contracting path. After each down sampling block, max-pooling layers were used, and the number of feature channels was doubled. An expanding path consisted of convolution and up sampling operations with long skip connections to integrate features from the corresponding down sampling operations. Each convolutional operation used a $3 \times 3 \times 3$ kernel size followed by a batch normalization and a rectified linear unit (ReLU) activation function.

In each epoch, 16 randomly positioned $64 \times 64 \times 64$ 4D patches (by concatenating intensity values from all 4 protocols used) were extracted from each image and label and fed into the network. This ensured that each epoch used a slightly different set of training data in addition to the data shuffling from the 5-fold cross-validation. The learning rate was set to 0.01, batch size was set to 8, and the number of epochs to 50. Stochastic gradient descent with a momentum optimization with an L_2 regularization factor was used. The DenseNet used a generalized dice loss function implemented through the dice pixel classification layer. For the segmentation evaluation, the tumor subregions from DenseNet output, NCR/NET-label 1, ED-label 2, and ET-label 4 were combined to create three segmentation labels namely whole tumor (WT-NCR/NET-label 1, ED-label 2, and ET-label 4), tumor core (TC-NCR/NET-label 1 and ET-label 4) and enhancing tumor (ET-label 4). All experiments were implemented in MATLAB on a workstation with an NVIDIA GeForce RTX 8000 with 48 GB RAM.

2.3 Extraction of image quality measurements from MRI scans

All scans from both BraTS 2020 and BraTS 2021 cohorts were evaluated via MRQy[21], an open-source quality control tool for extracting IQMs from large-scale MR imaging cohorts. MRQy works by first automatically detecting the foreground in MR images, followed by extraction of imaging-specific metadata (n = 10) as well as calculation of 13 IQMs per scan. In this study, we considered 10 IQMs for our study as shown in Table 1.

2.4 Experimental design

DenseNet performance was evaluated in terms of a 5-fold cross-validated dice coefficient with a random train-test split, which was considered the baseline performance. To evaluate the relationship between MR image quality and automated brain tumor segmentation accuracy, only the IQMs retained that exhibited a correlation with the baseline cross-validated dice coefficient of the DenseNet in the training cohort, as determined via Pearson correlation coefficient > 0.1. For each retained IQM, MRI scans from the BraTS 2020 training cohort were grouped as “better” quality (BQ) and “worse” quality (WQ) images based on the definitions in Table 1. This quality grouping was done separately based on IQMs of individual modalities, which showed the highest correlations with Dice. Since there is no definite cutoff value to threshold the better or worse quality images, a relative

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measurement of IQM values within the training pool images from BraTS 2020 was utilized instead. The training sub-cohorts were evaluated under two scenarios for each IQM: 1) BQtrain model, where the training set had better quality images and the test set had worse quality images; 2) WQtrain model, where the training set had worse quality images and the test set comprised better quality images. For example, as shown in Figure 1a for the CV-BQtrain model, 80% of images with low CV values (better quality images) were pooled into the training set and the remaining 20% of images (worse quality images) were considered as test set and vice versa for the CV-WQtrain model.

In every scenario, DenseNet performance was quantified in terms of dice coefficient for the WT region while training and test set ratios (80:20) were adjusted to be the same as a 5-fold cross-validation train-test split. For all the studies WT dice is taken for comparison since it includes all three tumor subregions. DenseNet performance in each scenario (BQtrain model, WQtrain model) was further evaluated in hold-out validation on the BraTS 2021 cohort, based on defined thresholds for selected IQMs. This yielded two defined hold-out test sets: (i) Mixed quality (MQ) images within the IQM range of both the BQtrain model and WQtrain model (Test set-1) and (ii) BQ images for the WQtrain model and WQ images for the BQtrain model (Test set-2). The overall experimental design is illustrated in Figure 1b.

2.5 Model assessment

The performance of the DenseNet model was evaluated with the dice similarity coefficient (DSC) as well as Hausdorff distance (HD). Dice similarity coefficient[25] between the predicted mask (P) and its corresponding ground truth mask (R) was calculated using $D=2\cdot|P\cap R| / (|P|+|R|)$, where P is the set of voxels in the predicted mask, R is the set of voxels in the ground truth mask. Hausdorff distance is calculated with simpleITK[26] by computing the distance between the set of non-zero pixels of two images (predicted mask and ground truth mask). In this work, this was adapted to yield Hausdorff95, a metric to measure the 95% quantile of the surface distance. Note that all evaluation measures only considered the WT region as defined above, as this was an aggregation of publicly available NCR/NET-label 1, ED-label 2, and ET-label 4 subregions.

3 RESULTS

3.1 Correlation between IQMs and segmentation performance

The 3D-DenseNet model achieved dice values of 0.7280, 0.7014, and 0.6726 for the whole tumor, tumor core, and enhancing tumor, respectively in a 5-fold cross-validation on the training set. Figures 2 and 3 show bar plots depicting the relationship (via Pearson correlation coefficient) between IQMs of all four image modalities (T1, T2, T1Gd, and T2-FLAIR) and cross-validated WT dice & ET dice, respectively. IQMs of T2-FLAIR images show a higher correlation with WT dice compared to other modalities (Figure 2). Additionally, IQMs of T1Gd images showed a better correlation with ET dice (Figure 3). We also ran additional trials (a total of 10 trials) with the random cohort selection method to confirm that we are indeed seeing a true signal from IQM-based training independent

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of the data sampling (Supporting information, Figure S-1). Dice scores from Trial 1 were considered for comparison with IQM-based training throughout the study.

3.2 Comparison of T2-FLAIR cohorts IQM-based training scenarios

Seven IQMs (CV, PSNR, SNR1, SNR2, SNR4, CNR, CJV) from T2-FLAIR images showed moderate correlation (>0.1) with DenseNet WT dice segmentation performance (Figure 2) and were selected for further analysis in this experiment. Figure 4 shows a significant improvement in the dice scores and Hausdorff distance (Supporting information, Figure S-2) when considering the BQtrain model based on CV and CJV, and for the WQtrain model based on PSNR, SNR4, and CNR; compared to baseline cross-validated performance (without considering IQM-segregating training sets). The overall highest dice scores were obtained for CV-BQtrain and CJV-BQtrain models. These trends are further evaluated in terms of WT dice scores in two hold-out validation test sets as summarized in Table 2. For test set-1 (MQ images), Model CV-BQtrain yielded the best overall WT tumor segmentation accuracy. Additionally, dice scores were improved when considering BQtrain models for CV, SNR2, SNR4, and CJV or when considering WQtrain models for PSNR, SNR1, or CNR; compared to their inverse models. For test set-2, dice scores were better for BQtrain models based on CV, CJV, and WQtrain models based on PSNR, SNR1, SNR2, SNR4 and CNR compared to their inverse models.

3.3 Comparison of T1Gd cohorts IQM-based training scenarios

Based on the Pearson correlation coefficient (Figure 3), six IQMs (PSNR, SNR1, SNR2, SNR4, CNR, CVP) of T1Gd images were chosen for further analysis in this experiment. When compared to baseline cross-validated performance (without considering IQM-segregating training sets), there is an improvement in dice scores (Figure 5) and Hausdorff distance (Supporting information, Figure S-3) when considering the BQtrain model based on CVP, and for the WQtrain model based on PSNR, SNR1, SNR2, SNR4, and CNR. The overall highest dice scores were obtained for the PSNR-WQtrain and SNR4-WQtrain models. These models were further evaluated with two hold-out validation test sets and the WT dice scores are summarized in Table 3. For test set-1, no significant difference was observed in dice scores between the BQtrain and WQtrain models (Table 3), except for CVP-based segregation. WQtrain models based on PSNR, SNR, CNR, and BQtrain model based on CVP showed better segmentation accuracy in test set-2 compared to their inverse models.

4 DISCUSSIONS

In this study, we sought to quantitatively evaluate the relationship between specific MR image quality measures and DenseNet-based tumor segmentation performance using brain MRIs. To our knowledge, there have not been any large-scale studies of the relationship between MR image quality metrics and the performance of deep learning segmentation models. Our results could help inform how to improve the performance of automated tumor segmentation models and increase their generalizability on unseen multi-institutional cohorts. Fundamentally, the sensitivity analysis of a neural network is a difficult undertaking. Neural networks are commonly treated as a black box with a high dimensional

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parameter space. Systematically varying each parameter within a high-dimensional neural network to study the effect on the output can be computationally infeasible. Strategies that reduce the network's size to facilitate a practical study of parameter sensitivity may also be used[27], but it is not clear that results on a smaller network generalize to real-world applications.

In general, segmentation accuracy depends on the training data which should have the right representative of the test set. However, combining data from different centers for model training alone does not yield high accuracy. For the prostate segmentation, a network trained on one or more sites overestimated the segmentation accuracy for unseen site[16]. Site-specific acquisition protocol differences in multi-site data often impose challenges in model training due to the inter-site heterogeneity. Styner et al[28] studied the intra and inter-site variability by evaluating the coefficients of variability (COV). COVs were larger between the sites and between the sites of different scanner types. It is important to understand the inherent variability within the training cohort to overcome the inter-site heterogeneity and to develop a generalized model. Our approach considered model sensitivity analysis within the constrained aspect of MR image quality specifically. This was done via popularly utilized and well-defined IQMs that attempt to characterize and quantify specific aspects of image quality and acquisition artifacts. As a first step, in this paper, we demonstrated the impact of better quality (BQ) vs worse quality (WQ) images on neural network model training. Considering BQ/WQ images based on IQM influences the model training differently rather than taking the full spectrum of images.

We developed training sets based on IQMs from T2-FLAIR and T1Gd MRI scans. When considering mixed-quality images for testing (test set-1), both WQtrain and BQtrain models may be expected to yield similar segmentation accuracy. However, when compared to their inverse models, CV-BQtrain and CNR-WQtrain, based on T2-FLAIR IQMs (Table 2), and the model CVP-BQtrain based on T1Gd IQMs (Table 3) exhibited a significant increase (>0.1) in segmentation performance. This indicates that the overall quality of the images in the training pool does influence network learning efficiency and segmentation. For all other IQMs, only a marginal difference was observed between the BQtrain and WQtrain models when tested on mixed-quality images, suggesting a weaker relationship with model performance. For test set-2 with unseen cohorts, the models trained with BQ images based on inhomogeneity measurements for IQMs CV, CJV, and CVP and the models trained with WQ images based on noise measurements for IQMs PSNR, SNR, and CNR yielded better segmentation accuracy and outperformed their inverse models. This shows that one can optimize the model performance by selecting the training images based on the IQM measures which can improve the segmentation accuracy significantly on unseen data.

Overall, in our experiments, models trained with better quality images as defined by IQMs of inhomogeneity artifacts (CV, CJV, CVP) yielded better segmentation accuracy. Studies have shown that smaller CJV measures in MR images (which corresponded to better quality in our experiments) indicate increased contrast between tissue regions[29], while higher CJV values (i.e., worse quality) correspond to the presence of heavy head motion and large artifacts[21]. Intuitively, the CJV-WQtrain model showed reduced overall segmentation performance compared to the CJV-BQtrain model. This implies that the high percentage

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of poor-quality images in the training pool did negatively impact segmentation accuracy. Correspondingly, the learning efficiency was effectively improved when considering the CJV-BQtrain model. Similar trends were observed for CV and CVP. In contrast, models trained with worse quality images (WQtrain models) as defined by IQMs of noise artifacts (PSNR, SNR, CNR) yielded better overall segmentation accuracy. When the model was trained on images only with high SNR (BQ images), segmentation accuracy was poor for low SNR images (WQ images). Including the low SNR images in the training pool (WQtrain model) improves the learning efficiency of the neural network and gives better segmentation accuracy for both WQ and BQ images.

We acknowledge some limitations in this study. We grouped training cohorts based on the IQM of individual modalities though the segmentation models utilized multi-modal input. It may be more beneficial to combine image quality across the protocols for training a multi-modal network. Our study focused exclusively on brain tumor segmentation and reproducing our findings on other segmentation problems with multi-site data will be valuable.

5 CONCLUSIONS

In this work, we conducted an initial evaluation of the relationship between MR image quality measurements and DenseNet segmentation performance using multi-parametric MRIs. Our results suggest a significant relationship may exist between specific MR image quality measures and DenseNet-based tumor segmentation performance. Training on BQ images based on inhomogeneity measurements and training on WQ images based on noise measurements yielded better segmentation accuracy regardless of the test image quality compared to their inverse models. Identifying the relative quality difference in the MRI cohorts based on IQM measures will help in the development of more accurate and generalizable models. This suggests that in addition to the image pre-processing to correct for artifacts, with the prior knowledge of IQM measures of the images, models can be generalized by selecting MRI scans based on IQM measures. This way multi-center data can be pipelined to the suitable model with the minimum required pre-processing prior to applying a segmentation model.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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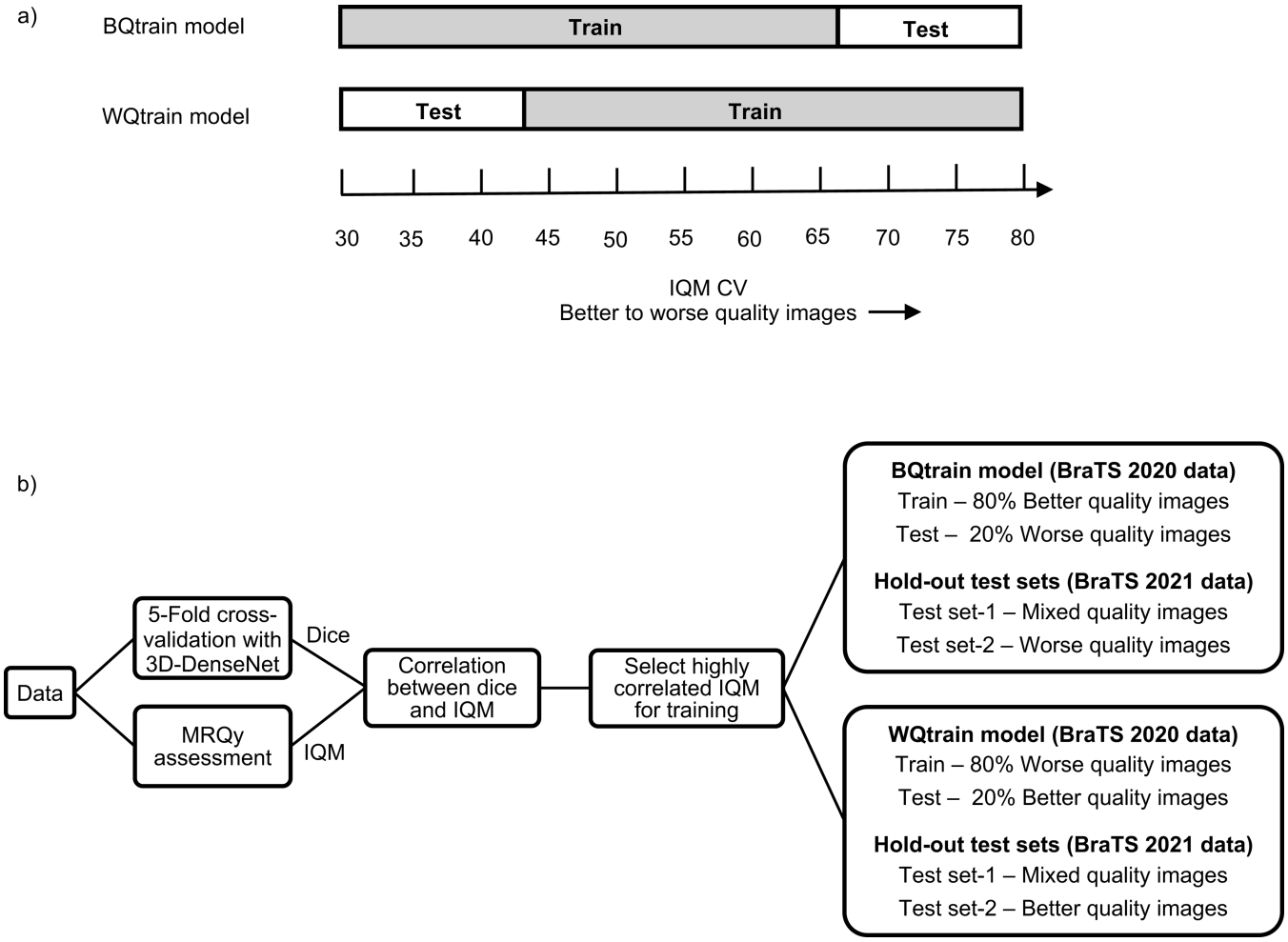


Figure 1.

a) A schematic diagram showing the selection of MRI scans for BQtrain and WQtrain models based on IQM CV. b) An overview of IQM-based model training.

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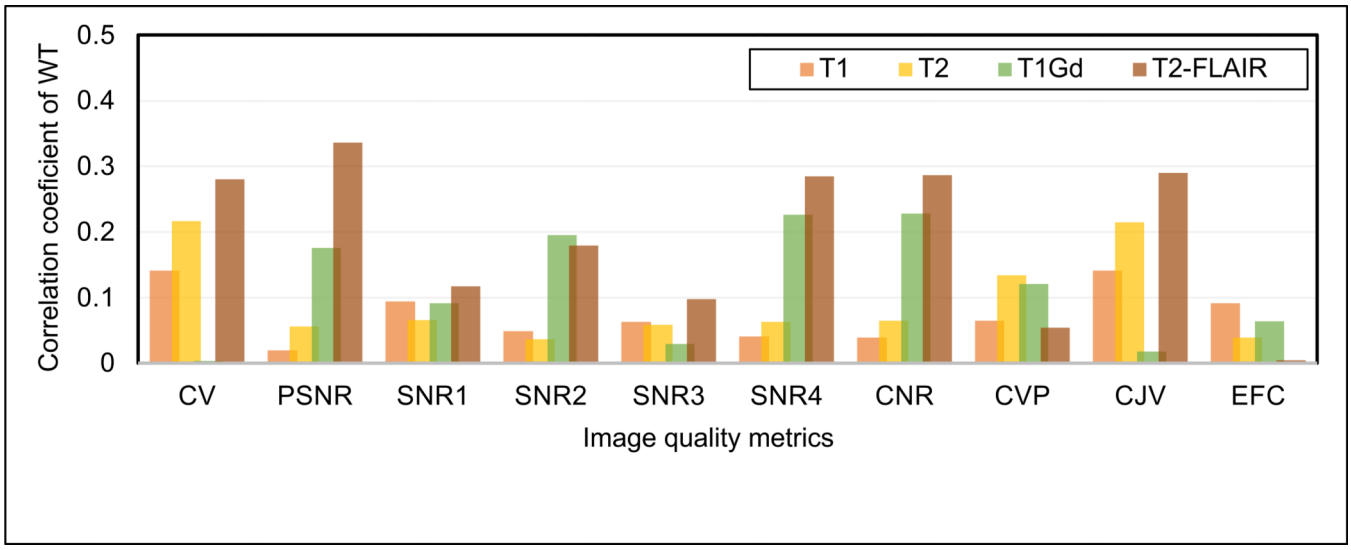


Figure 2.

Bar plot of the correlation coefficient between cross-validated whole tumor (WT) dice with the IQM measures from each modality in the BraTS 2020 training cohort.

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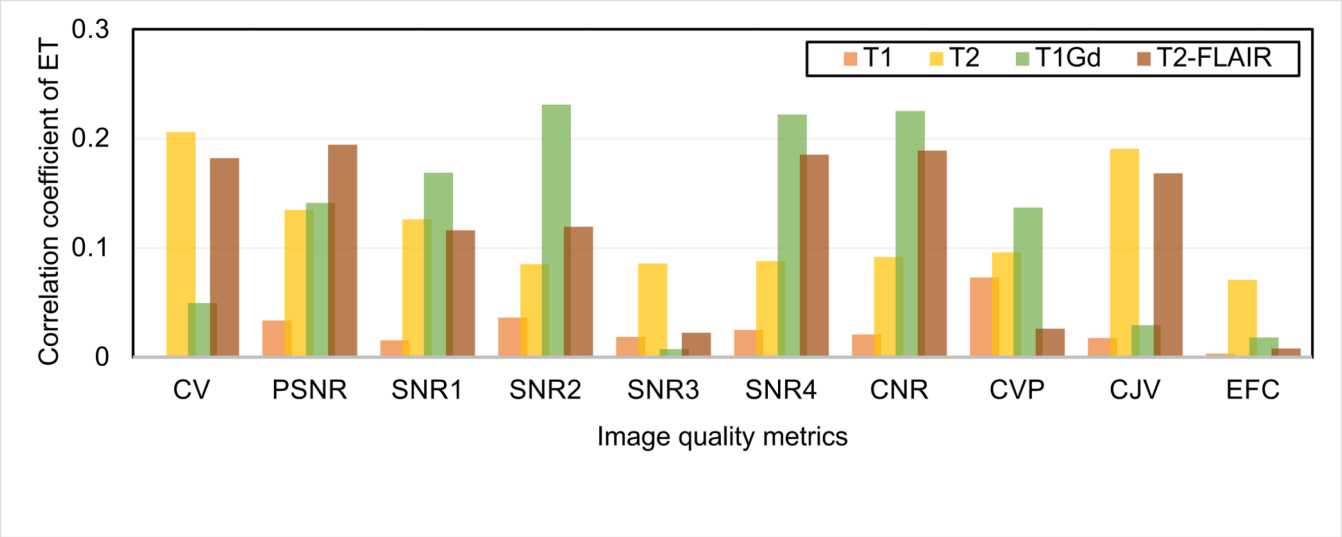


Figure 3.

Bar plot of the correlation coefficient between cross-validated enhancing tumor (ET) dice with the IQM measures from each modality in the BraTS 2020 training cohort.

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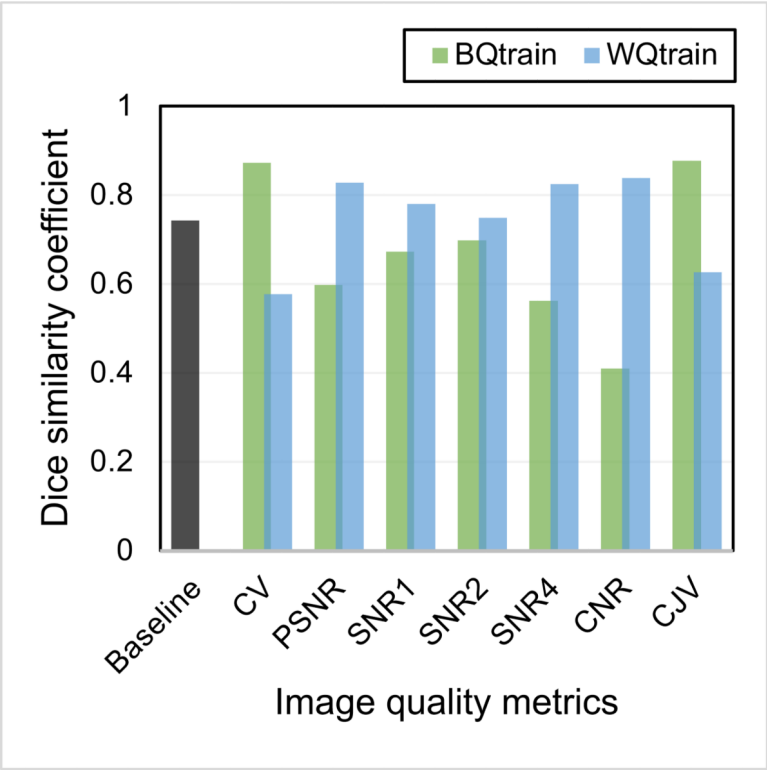


Figure 4.

Mean dice similarity coefficient for the whole tumor (WT) when using BQtrain and WQtrain models as defined by each selected IQM measure from T2-FLAIR, BraTS 2020 training cohort.

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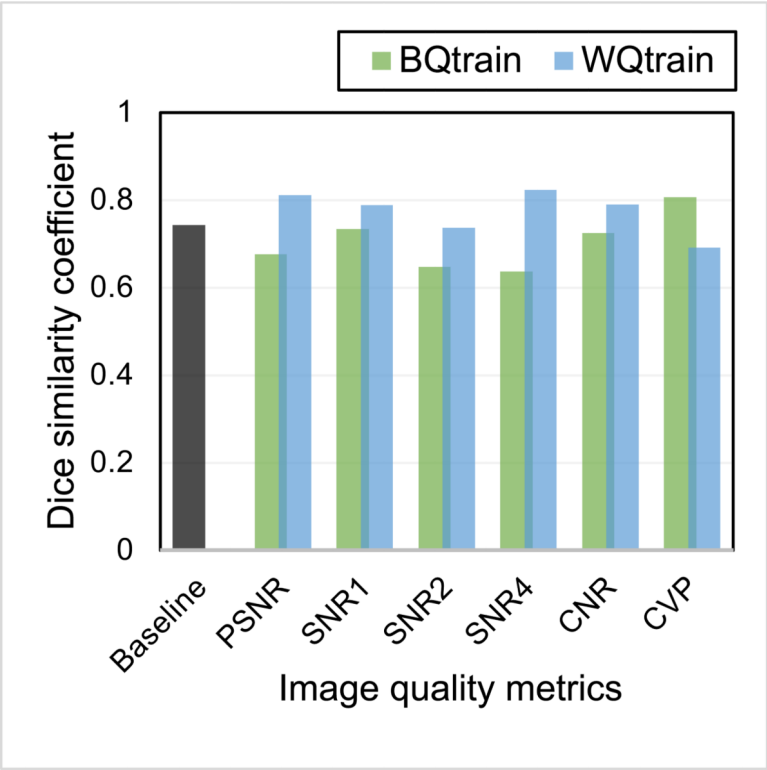


Figure 5.
Mean dice similarity coefficient for the whole tumor (WT) when using BQtrain and WQtrain models as defined by each selected IQM measure from T1Gd, BraTS 2020 training cohort.

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Table 1:

IQM definitions and approach to segregate images into better and worse quality sub-cohorts.

Image quality metrics (IQM)	Reference for better quality (BQ) images[21]
Coefficient of variation of the foreground for shadowing and inhomogeneity artifacts (CV)	Smaller values
Peak signal to noise ratio of the foreground (PSNR)	Larger values
Signal to noise ratio, foreground standard deviation (SD) divided by background SD (SNR1)	Larger values
Mean of the foreground patch divided by background SD (SNR2)	Larger values
Foreground patch SD divided by the centered foreground patch SD (SNR3)	Larger values
Mean of the foreground patch divided by mean of the background patch (SNR4)	Larger values
Contrast to noise ratio for shadowing and noise artifacts, mean of the foreground and background patches difference divided by background patch SD (CNR)	Larger values
Coefficient of variation of the foreground patch for shading artifacts: foreground patch SD divided by foreground patch mean (CVP)	Smaller values
Coefficient of joint variation between the foreground and background for aliasing and inhomogeneity artifacts (CJV)	Smaller values
Entropy focus criterion for motion artifacts (EFC)	Smaller values

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Table 2:

WT dice scores for BQtrain and WQtrain models (based on IQMs from T2-FLAIR images) when evaluated in hold-out setting on MQ, BQ, and WQ images from the BraTS 2021 cohort.

BraTS 2020 dataset		BraTS 2021 dataset			
Training based on IQM measure of T2-FLAIR images (306 images)		Test set-1 (70 MQ images)		Test set-2 (70 BQ/WQ images)	
Models	IQM range	IQM range	WT dice	IQM range	WT dice
CV-BQtrain	30.84–50.96	40–50	0.8348	51–79	0.8589
CV-WQtrain	40.65–62.94		0.7354	31–40	0.5627
PSNR-BQtrain	13.44–20.18	13.44–17.16	0.6930	11.54–13.3	0.6354
PSNR-WQtrain	11.20–17.16		0.7484	17.2–20.75	0.8525
SNR1-BQtrain	34.45–186.59	34.45–62.23	0.7365	16.09–34.30	0.5943
SNR1-WQtrain	15.96–62.23		0.7892	62.3–124.14	0.7913
SNR2-BQtrain	117.02–610.75	117.02–230.68	0.8244	49.60–116	0.7020
SNR2-WQtrain	62.83–230.68		0.7685	231–481.82	0.8014
SNR4-BQtrain	9.00–37.20	9.00–12.25	0.7889	6.24–8.9	0.6033
SNR4-WQtrain	6.22–12.25		0.7666	12.3–18.41	0.8737
CNR-BQtrain	8.54–36.78	8.54–11.84	0.5787	5.7–8.4	0.4479
CNR-WQtrain	5.75–11.84		0.6890	11.9–18.0	0.8814
CJV-BQtrain	0.32–0.51	0.41–051	0.7717	0.52–0.81	0.8230
CJV-WQtrain	0.41–0.64		0.7104	0.32–0.40	0.5535

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Table 3:

WT dice scores for BQtrain and WQtrain models (based on IQMs from T1Gd images) when evaluated in hold-out setting on MQ, BQ, and WQ images from the BraTS 2021 cohort.

BraTS 2020 dataset		BraTS 2021 dataset			
Training based on IQM measure of T1Gd images (306 images)		Test set-1 (70 MQ images)		Test set-2 (70 BQ/WQ images)	
Models	**IQM Range**	**IQM Range**	**WT Dice**	**IQM Range**	**WT Dice**
PSNR-BQtrain	15.24–21.8	15.24–19.62	0.6999	12.89–15.20	0.6366
PSNR-WQtrain	12.82–19.62		0.7128	19.63–23.34	0.7818
SNR1-BQtrain	48.92–181.25	48.92–112.38	0.8545	24.69–48.88	0.7231
SNR1-WQtrain	15.69–112.38		0.8190	112.5–159.26	0.7947
SNR2-BQtrain	192.88–841.09	192.88–493.61	0.7565	89.36–193.0	0.6996
SNR2-WQtrain	83.06–493.61		0.7477	494.0–899.20	0.7796
SNR4-BQtrain	10.78–23.99	10.78–16.25	0.7652	7.88–10.75	0.6460
SNR4-WQtrain	6.82–16.25		0.7648	16.26–29.61	0.7922
CNR-BQtrain	10.38–23.74	10.38–15.98	0.8160	7.38–10.37	0.7161
CNR-WQtrain	6.37–15.98		0.7703	16.0–29.27	0.7436
CVP-BQtrain	0.22–0.751	0.53–0.75	0.8331	0.76–1.01	0.8146
CVP-WQtrain	0.531–0.95		0.7220	0.26–0.52	0.7248